

**Notes: Chen, Cui, He, and Milbradt (RFS, 2018)**  
**Quantifying Liquidity and Default Risks of Corporate Bonds over the Business Cycle**

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# 1 Big picture

- **Question.** How much of corporate bond spreads reflects default risk, how much reflects liquidity frictions, and how important is the *interaction* between the two?
- **Setting.** The paper studies U.S. corporate bonds in an environment with business-cycle risk, endogenous default, debt rollover, and over-the-counter secondary-market trading frictions.
- **Core challenge.** In the data, credit spreads, bid-ask spreads, and bond-CDS spreads comove with ratings and the business cycle. Reduced-form decompositions usually add default and liquidity components, but in reality default and liquidity move together: lower liquidity can raise default risk, and higher default risk can worsen liquidity.
- **Main model.** The authors build a structural model in which firms face rollover risk, investors face idiosyncratic liquidity shocks, dealers intermediate trade in an OTC market, and investors' holding costs rise as bond prices fall. This creates a *default-liquidity spiral*.
- **Main quantitative result.** The model matches average default rates and credit spreads across ratings, and also generates realistic bid-ask spreads and bond-CDS spreads over the business cycle.
- **Key decomposition result.** Default-liquidity interactions account for **10%–24%** of the level of credit spreads in the cross section and **16%–46%** of the business-cycle variation in spreads.
- **Aggregate financing-cost result.** Over 1996–2015, the paper estimates about **\$2.9 trillion** of bond-value losses for new corporate issuance due to default and liquidity frictions together, about **14%** of total issuance. The liquidity-related components account for about **43%** of these losses.
- **Policy result.** A policy that improves recession-time secondary-market liquidity to normal-time conditions lowers Ba spreads in recession by about **102 bps** and raises the aggregate value of the U.S. corporate bond market by about **\$256 billion**. Importantly, much of this effect comes from the endogenous feedback between default and liquidity rather than from the direct liquidity effect alone.
- **Economic takeaway.** Secondary-market liquidity is not just a “pricing residual.” It changes firms' default incentives through rollover risk, and this feedback materially affects credit spreads, financing costs, and the value of liquidity interventions.

## 2 Data and measurement (key objects)

### 2.1 Observed data and empirical moments

- **Default rates.** Cumulative default probabilities by rating and maturity come from *Moody's* default studies (1920–2011).
- **Bond credit spreads.** Transaction-level corporate bond data come from *Mergent FISD* (1994–2004) and *TRACE* (2005–2012). The sample excludes utilities and financial firms. Credit spreads are bond yields minus Treasury yields of corresponding maturity.
- **Bid-ask spreads.** Normal-time bid-ask spreads by rating are taken from *Edwards, Harris, and Piwowar (2007)*. Recession bid-ask spreads are constructed by scaling these normal-time

numbers using the recession/expansion ratio from Roll-style illiquidity measures following *Bao, Pan, and Wang (2011)*.

- **Bond–CDS spreads.** CDS data come from *Markit* starting in 2005. The bond–CDS spread is bond credit spread minus CDS spread and is interpreted as an additional measure of bond illiquidity.
- **Leverage data.** Firm leverage distributions come from *Compustat* from 1994 to 2012. The model uses observed leverage distributions within rating classes to aggregate firm-level model moments.
- **Aggregate issuance data.** *SIFMA* data on annual U.S. corporate bond issuance (1996–2015) are used to turn spread decompositions into dollar financing-cost decompositions.

## 2.2 Aggregate states, bonds, and ratings

- **Aggregate states.** The model uses two aggregate states:
  - $G$  = normal/expansion state,
  - $B$  = recession/distress state.

In the empirical implementation, NBER recessions define the bad state.

- **Financial-crisis exclusion.** The quarters **2008Q4** and **2009Q1** are excluded from the baseline calibration because the paper wants the bad state to represent a *normal recession*, not the extraordinary disruption of the global financial crisis.
- **Rating classes.** The paper works with four ratings: **Aaa/Aa**, **A**, **Baa**, and **Ba**. The first three are investment-grade, and Ba is speculative-grade.
- **Maturity focus.** The quantitative analysis targets **10-year bonds**, although the paper also reports 5-year default probabilities and credit spreads. The focus on 10-year maturity is motivated by the average maturity of corporate bond issuance and by the existing structural credit literature.

## 2.3 Key target moments and benchmark numbers

- **10-year cumulative default probabilities (data vs. model).**
  - Aaa/Aa: 2.1% vs. 1.6%
  - A: 3.4% vs. 3.9%
  - Baa: 7.0% vs. 7.9%
  - Ba: 19.0% vs. 15.9%
- **10-year credit spreads in state  $G$  (data vs. model, bps).**
  - Aaa/Aa: 61 vs. 86
  - A: 90 vs. 122
  - Baa: 150 vs. 182
  - Ba: 303 vs. 301

- **10-year credit spreads in state  $B$  (data vs. model, bps).**
  - Aaa/Aa: 106 vs. 136
  - A: 159 vs. 185
  - Baa: 262 vs. 261
  - Ba: 454 vs. 404
- **Bid-ask spreads (data vs. model, bps).**
  - State  $G$ : superior 40 vs. 39, investment 50 vs. 44, junk 70 vs. 61
  - State  $B$ : superior 77 vs. 111, investment 125 vs. 128, junk 218 vs. 186
- **Bond–CDS spreads (data vs. model, bps).**
  - State  $G$ : Aaa/Aa 23 vs. 48, A 37 vs. 53, Baa 58 vs. 61, Ba 68 vs. 61
  - State  $B$ : Aaa/Aa 72 vs. 69, A 104 vs. 79, Baa 162 vs. 92, Ba 191 vs. 107

## 2.4 Unobservables and timing

- **Aggregate uncertainty.** Firms face macroeconomic state transitions, time-varying aggregate risk prices, and recession-dependent market liquidity.
- **Firm cash-flow risk.** Each firm has systematic and idiosyncratic cash-flow risk. Firms differ in leverage, which is the key cross-sectional state variable in the quantitative mapping.
- **Investor liquidity type.** Bondholders can be in a high-liquidity state  $H$  or a low-liquidity state  $L$ . They are hit by idiosyncratic liquidity shocks that make them costly holders of the bond.
- **OTC trading friction.** L-type investors must locate dealers and then buyers through a search process. This creates bid prices, ask prices, and bid-ask spreads endogenously.
- **Rollover timing.** Corporate debt has a smooth maturity structure: a fraction of outstanding debt matures each instant, and the firm must refinance at current market conditions. This is the channel through which secondary-market liquidity affects default incentives.

## 3 Models and empirical specifications (all equations + interpretation)

### 3.1 Aggregate states and the stochastic discount factor

The aggregate state follows a continuous-time Markov chain, with state  $s_t \in \{G, B\}$ . The stochastic discount factor is

$$\frac{dM_t}{M_t} = -r(s_t)dt - \eta(s_t)dZ_t^m + \sum_{s_t^- \neq s_t} (e^{\kappa(s_t^-, s_t)} - 1)dM_t^{(s_t^-, s_t)}. \quad (1)$$

- $r(s_t)$  is the state-dependent risk-free rate.
- $\eta(s_t)$  is the market price of aggregate Brownian risk.

- $\kappa(\cdot)$  governs the jump-risk premium attached to aggregate state switches.

*Interpretation.* The paper prices corporate bonds in an environment where both macroeconomic conditions and the price of aggregate risk vary over the business cycle. This matters because both default risk and liquidity premia are higher in recessions.

### 3.2 Firm cash flows, risk-neutral dynamics, and unlevered value

Firm cash flows  $Y_t$  follow

$$\frac{dY_t}{Y_t} = \mu^P(s_t)dt + \sigma_m(s_t)dZ_t^m + \sigma_f dZ_t^f. \quad (2)$$

Under the risk-neutral measure, log cash flow  $y_t = \log Y_t$  evolves as

$$dy_t = \mu_s dt + \sigma_s dZ_t^Q, \quad (3)$$

with

$$\mu_s \equiv \mu^P(s) - \sigma_m(s)\eta(s) - \frac{1}{2}(\sigma_m(s)^2 + \sigma_f^2), \quad \sigma_s \equiv \sqrt{\sigma_m(s)^2 + \sigma_f^2}.$$

The unlevered firm value is  $v_s^U Y$ , where the state-contingent price-dividend ratio vector solves

$$\begin{pmatrix} v_G^U \\ v_B^U \end{pmatrix} = \begin{pmatrix} r_G - \mu_G + \zeta_G & -\zeta_G \\ -\zeta_B & r_B - \mu_B + \zeta_B \end{pmatrix}^{-1} \begin{pmatrix} 1 \\ 1 \end{pmatrix}. \quad (4)$$

*Interpretation.* The model starts with a standard structural credit setup: firms generate stochastic cash flows whose present value depends on the macro state. What is new is that these cash flows later interact with rollover risk and secondary-market liquidity.

### 3.3 Debt maturity, rollover, and the key financing friction

The firm has a unit measure of outstanding bonds with coupon  $c$  and face value  $p$ . Each bond matures with intensity  $m$ , so the expected average maturity is  $1/m$ . Equityholders keep aggregate face value and coupon constant by refinancing maturing bonds with new bonds of the same average maturity.

- If  $m = 0.2$ , average maturity is 5 years.
- Higher  $m$  means more frequent refinancing and therefore more rollover exposure to current liquidity conditions.

*Interpretation.* This smooth-maturity structure makes the firm continuously exposed to conditions in the primary and secondary bond market. Liquidity frictions matter not because they directly change operating cash flows, but because they affect refinancing costs and thus the default boundary.

### 3.4 OTC bond market, liquidity shocks, and holding costs

Bondholders are either in a high-liquidity state  $H$  or low-liquidity state  $L$ . H-type investors hold the bond without a holding cost. They are hit by idiosyncratic liquidity shocks with intensity  $\xi_s$ , turning them into L-types who incur a holding cost

$$h_s^c(P^s(y)) = \chi_s[N - P^s(y)]. \quad (5)$$

- $P^s(y)$  is the endogenous mid-price of the bond.
- $N$  is an intercept in the holding-cost schedule.
- $\chi_s$  is the slope, allowed to vary by aggregate state.

This specification is micro-founded in the appendix through costly collateralized financing. When the bond price falls and/or haircuts rise, the investor must rely more on expensive uncollateralized financing, which raises the effective holding cost.

*Interpretation.* This is the central modeling innovation. Because holding costs increase as the bond price falls, and bond prices fall as default risk rises, default and liquidity become tightly linked rather than additively separable.

### 3.5 Dealers, bid and ask prices, and the mid-price

L-type sellers and H-type buyers must trade through dealers in an OTC market. Sellers meet dealers with intensity  $\lambda_s$  and buyers also meet dealers with intensity  $\lambda_s$ . Nash bargaining gives investors bargaining weight  $\beta$  and dealers weight  $1 - \beta$ .

Fixing investor valuations  $D_H^s$  and  $D_L^s$ , the equilibrium prices satisfy:

$$A^s = D_H^s, \quad B^s = \beta D_H^s + (1 - \beta) D_L^s.$$

Hence the dollar bid-ask spread is

$$A^s - B^s = (1 - \beta)(D_H^s - D_L^s).$$

The bond “market price” is defined as the mid-price:

$$P^s(y) = \frac{A^s(y) + B^s(y)}{2} = \frac{(1 + \beta)D_H^s(y) + (1 - \beta)D_L^s(y)}{2}. \quad (6)$$

The proportional bid-ask spread is

$$ba^s(y) = \frac{A^s(y) - B^s(y)}{P^s(y)} = \frac{2(1 - \beta)(D_H^s(y) - D_L^s(y))}{(1 + \beta)D_H^s(y) + (1 - \beta)D_L^s(y)}. \quad (7)$$

*Interpretation.* Bid-ask spreads measure the gap between H-type and L-type investor valuations. Bond–CDS spreads, by contrast, capture broader expected liquidity costs. The paper later uses comparative statics to show these are not the same object.

### 3.6 Bankruptcy, recovery, and endogenous default

Equityholders default when the option value of continuing the firm no longer covers coupon payments and rollover losses. Since default is more attractive in bad macro states, the bad-state default threshold lies above the good-state threshold:

$$y_{\text{def}}^G < y_{\text{def}}^B.$$

At default, recovery is type- and state-dependent:

$$D_{\text{def}}(y) = \begin{pmatrix} \alpha_H^G v_G^U \\ \alpha_L^G v_G^U \\ \alpha_H^B v_B^U \\ \alpha_L^B v_B^U \end{pmatrix} e^y. \quad (8)$$

The paper then solves for closed-form bond and equity values on intervals defined by the default boundaries. Schematically, bond values take the form

$$D^{(i)}(y) = G^{(i)} \exp(\Lambda^{(i)} y) b^{(i)} + k_0^{(i)} + e^y k_1^{(i)}, \quad (9)$$

and equity values take a similar exponential-affine form,

$$E^{(i)}(y) = (\text{state-contingent exponential terms}) + e^y \tilde{k}_1^{(i)}. \quad (10)$$

Optimal default boundaries satisfy smooth-pasting conditions:

$$E'_G(y_{\text{def}}^G) = 0, \quad E'_B(y_{\text{def}}^B) = 0. \quad (11)$$

*Interpretation.* Default is endogenous and forward-looking. Equityholders internalize that a worse secondary bond market raises future rollover losses, so the firm defaults earlier when liquidity worsens.

### 3.7 Calibration strategy

The benchmark calibration uses a combination of pre-fixed and calibrated parameters.

- **Pre-fixed macro/asset-pricing parameters.**

- Transition intensities:  $\zeta_G^P = 0.1$ ,  $\zeta_B^P = 0.5$
- Jump-risk premium factor:  $\exp(\kappa_G) = 2.0$ ,  $\exp(\kappa_B) = 0.5$
- Risk prices:  $\eta_G = 0.17$ ,  $\eta_B = 0.22$
- Risk-free rate:  $r = 5\%$
- Cash-flow growth:  $\mu_G^P = 4.5\%$ ,  $\mu_B^P = 1.5\%$
- Systematic volatility:  $\sigma_m^G = 0.10$ ,  $\sigma_m^B = 0.11$
- Issuance cost:  $\omega = 1\%$
- Maturity intensity:  $m = 0.2$
- Treasury liquidity premia: 15 bps in  $G$ , 40 bps in  $B$
- Meeting intensities:  $\lambda_G = 50$ ,  $\lambda_B = 20$
- Liquidity shock intensities:  $\xi_G = 0.7$ ,  $\xi_B = 1.0$
- Bargaining power:  $\beta = 0.05$

- **Calibrated parameters.**

- Idiosyncratic cash-flow volatility:  $\sigma_f = 0.25$



- Holding cost intercept:  $N = 115$
- Holding cost slopes:  $\chi_G = 0.06$ ,  $\chi_B = 0.11$

Identification logic is moment-based:

- $\sigma_f$  is pinned down by Baa default rates,
- $N$  and  $\chi_s$  are pinned down by bid-ask spreads across ratings and states,
- $\xi_s$  is disciplined by TRACE turnover,
- recovery parameters  $\alpha_l^s$  are disciplined by prices of defaulted bonds and bid-ask spreads.

*Interpretation.* The calibration is designed so that the model has to match *both* classic credit moments and newer liquidity moments. This is what makes the paper able to say something meaningful about the interaction between default and liquidity.

### 3.8 Structural spread decomposition

The paper removes the exogenous Treasury liquidity premium and decomposes the credit spread relative to the risk-free rate as

$$\widehat{cs}^{rf} = \underbrace{\widehat{cs}^{\text{pure DEF}} + \widehat{cs}^{\text{LIQ} \rightarrow \text{DEF}}}_{\text{Default component}} + \underbrace{\widehat{cs}^{\text{pure LIQ}} + \widehat{cs}^{\text{DEF} \rightarrow \text{LIQ}}}_{\text{Liquidity component}}. \quad (15)$$

The four pieces are:

- **Pure default:** default spread in a frictionless secondary market with the default boundary reoptimized as in Leland (1994).
- **Liquidity-driven default:** additional default spread caused by worse rollover conditions when the secondary market is illiquid.
- **Pure liquidity:** spread of a default-free bond subject only to OTC liquidity frictions.
- **Default-driven liquidity:** additional liquidity spread caused by the bond becoming riskier and thus more expensive to hold/trade.

*Interpretation.* The standard additive decomposition is nested as a special case where the two interaction terms are zero. The whole point of the paper is that they are not zero and can be quantitatively important.

### 3.9 Liquidity-provision counterfactual

The paper studies a stylized recession liquidity intervention that improves state- $B$  secondary-market liquidity to state- $G$  conditions:

$$\chi_B \rightarrow \chi_G, \quad \lambda_B \rightarrow \lambda_G.$$

Interpreted literally, the policy makes recession-time bond-market liquidity as good as normal-time liquidity, for example through central-bank backstops or collateralized lending facilities that improve dealers' willingness to intermediate trades.

*Interpretation.* This counterfactual is not meant to replicate a specific program one-for-one. Rather, it isolates how much a better secondary market matters once one allows default and liquidity to feed back on each other.

## 4 Identification and model credibility (what makes the quantitative exercise persuasive)

### 4.1 What each empirical moment pins down

- **Default rates** identify the amount of firm-side idiosyncratic cash-flow risk needed to generate realistic default frequencies.
- **Credit spreads across ratings and states** discipline how leverage, macro state, and risk pricing translate into bond premia.
- **Bid-ask spreads** identify the intensity of holding costs and how those costs vary with bond price and the aggregate state.
- **Bond–CDS spreads** provide an out-of-target check on whether the same liquidity mechanism that fits bid-ask spreads also fits broader liquidity premia.
- **Leverage distributions** are crucial because spreads are convex in leverage. The model does not match rating-class moments at the mean leverage; it integrates across the observed cross section of leverage within each rating.

### 4.2 Why price-dependent holding costs are crucial

- A version of the model with constant state-dependent holding costs but *no* price dependence can be recalibrated to fit average bid-ask spreads, but it fails to generate the strong cross-sectional differences in liquidity across ratings.
- In that alternative model, lower-rated bonds are not much more illiquid than safer bonds, so the model also has trouble matching the cross-sectional pattern of credit spreads and bond–CDS spreads.
- This is the empirical value of the paper’s main mechanism: if holding costs do not worsen as a bond approaches default, then the model loses the default–liquidity spiral that is needed to explain the data.

### 4.3 Why bid-ask spreads and bond–CDS spreads are not redundant

- Bid-ask spreads reflect the valuation gap between H-type and L-type investors at the moment of trade.
- Bond–CDS spreads reflect the overall severity of liquidity frictions priced into the bond, including expected holding costs and the effect of those costs on rollover risk.
- Comparative statics show these two objects can move differently. For example, higher liquidity-shock intensity can raise expected holding costs and bond–CDS spreads while lowering bid-ask spreads by bringing H- and L-type valuations closer together.

## 4.4 Interpretation and limitations

- The macro environment has only two aggregate states, so the model is not designed to match extreme crisis episodes like late 2008 exactly.
- The paper treats the stochastic discount factor as exogenous.
- Firms differ mainly through leverage, not through richer heterogeneity in operations or cash holdings.
- The CDS market is treated as effectively frictionless in the baseline interpretation, which is useful but imperfect empirically.
- The analysis focuses mainly on 10-year bonds, leaving the full term structure challenge largely aside.

## 4.5 Relation to the literature

- The paper extends *He and Milbradt (2014)* by adding macroeconomic dynamics and bond-price-dependent holding costs.
- It speaks to the *credit spread puzzle*: standard structural models can fit default risk better once macro risk is added, but they remain largely silent on nondefault components. This paper jointly studies default and liquidity.
- It also contributes methodologically by offering a structural decomposition of credit spreads that goes beyond the additive default-plus-liquidity view common in reduced-form work.

# 5 Tables and figures

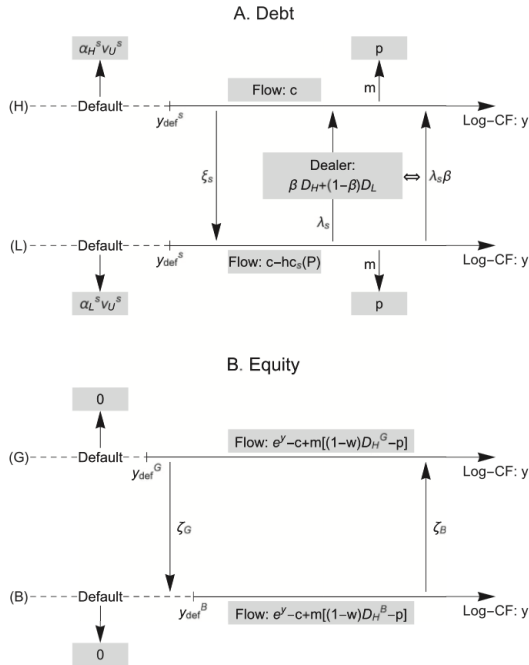
## 5.1 Figure 1: cash flows to debt and equity holders

**Read:**

- Figure 1 is the map of the whole model. Panel A tracks cash flows to bondholders in high- and low-liquidity states. Panel B tracks cash flows to equityholders, including coupon payments, refinancing proceeds, and state switches.
- The figure makes the logic of the default–liquidity spiral very intuitive: bad secondary-market liquidity lowers the price at which new debt can be issued, which hurts equityholders through rollover losses and can trigger earlier default.
- This figure is especially important because the mechanism is dynamic: the secondary market matters for the primary market because new buyers know they may become illiquid sellers in the future.

## 5.2 Table 1: benchmark parameters

**Read:**

Table 1  
Benchmark parameters

| Symbol                          | Description                  | State G | State B | Justification / Target                                             |
|---------------------------------|------------------------------|---------|---------|--------------------------------------------------------------------|
| <i>A. Pre-fixed parameters</i>  |                              |         |         |                                                                    |
| $\kappa^P$                      | Transition density           | 0.1     | 0.5     | <a href="#">Chen 2010</a>                                          |
| $\exp(\kappa)$                  | Jump risk premium            | 2.0     | 0.5     | <a href="#">Chen 2010</a>                                          |
| $\eta$                          | Risk price                   | 0.17    | 0.22    | <a href="#">Chen 2010</a>                                          |
| $r$                             | Risk-free rate               | 0.05    |         |                                                                    |
| $\mu_P$                         | Cash flow growth             | 0.045   | 0.015   | Corporate profit data                                              |
| $\sigma_P$                      | Systematic vol               | 0.10    | 0.11    | Equity volatility                                                  |
| $\alpha$                        | Primary market issuance cost | 0.10    | 0.01    | <a href="#">Chen 2010</a>                                          |
| $m$                             | Average maturity intensity   | 0.2     |         |                                                                    |
| $\Delta$                        | Treasury liquidity premium   | 15 bps  | 40 bps  | <a href="#">Repo-Treasury spread</a>                               |
| $\lambda$                       | Meeting intensity            | 50      | 20      | Anecdotal evidence                                                 |
| $\xi$                           | Liquidity risk intensity     | 0.7     | 1.0     | <a href="#">Bond turnover rate (TRACE)</a>                         |
| $\beta$                         | Investor's bargaining power  | 0.05    |         |                                                                    |
| $\alpha_H$                      | Recovery rate of $H$ type    | 58.71%  | 32.56%  | <a href="#">Bid prices for defaulted bonds and bid-ask spreads</a> |
| $\alpha_L$                      | Recovery rate of $L$ type    | 57.49%  | 30.50%  | <a href="#">Bid prices for defaulted bonds and bid-ask spreads</a> |
| <i>B. Calibrated parameters</i> |                              |         |         |                                                                    |
| $\sigma_f$                      | Idiosyncratic vol            | 0.25    |         |                                                                    |
| $N$                             | Holding cost intercept       | 115     |         |                                                                    |
| $\chi$                          | Holding cost slope           | 0.06    | 0.11    | <a href="#">Investment grade bid-ask spreads (G and B)</a>         |
|                                 |                              |         |         | <a href="#">Superior bid-ask spread in state G</a>                 |

- Table 1 separates pre-fixed parameters from calibrated ones. The model is not trying to estimate everything from a few moments; some parameters are disciplined from outside evidence or standard macro-finance targets.
- The key calibrated liquidity parameters are surprisingly parsimonious: one intercept  $N$  and two slopes  $\chi_G, \chi_B$ . Yet these are enough, combined with the structure of the model, to generate realistic cross-sectional and cyclical liquidity differences.
- The state- $B$  parameters always imply worse market functioning than state  $G$ : higher liquidity-shock intensity, lower meeting intensity, larger Treasury convenience yield, and a steeper holding-cost slope.

Table 2

Default probabilities and credit spreads across credit ratings

|                            | Maturity = 5 years |      |      |      | Maturity = 10 years |      |      |      |
|----------------------------|--------------------|------|------|------|---------------------|------|------|------|
|                            | Aaa/Aa             | A    | Baa  | Ba   | Aaa/Aa              | A    | Baa  | Ba   |
| A. Default probability (%) |                    |      |      |      |                     |      |      |      |
| Data                       | 0.7                | 1.3  | 3.1  | 9.8  | 2.1                 | 3.4  | 7.0  | 19.0 |
| Model                      | 0.3                | 0.8  | 2.4  | 7.4  | 1.6                 | 3.9  | 7.9  | 15.9 |
| B. Credit spreads (bps)    |                    |      |      |      |                     |      |      |      |
|                            | State G            |      |      |      | State B             |      |      |      |
|                            | Aaa/Aa             | A    | Baa  | Ba   | Aaa/Aa              | A    | Baa  | Ba   |
| Data                       | 56                 | 86   | 149  | 315  | 61                  | 90   | 150  | 303  |
|                            | (4)                | (7)  | (15) | (34) | (4)                 | (6)  | (13) | (23) |
| Model                      | 58                 | 73   | 114  | 237  | 86                  | 122  | 182  | 301  |
|                            | State B            |      |      |      | State G             |      |      |      |
|                            | Aaa/Aa             | A    | Baa  | Ba   | Aaa/Aa              | A    | Baa  | Ba   |
| Data                       | 107                | 171  | 275  | 542  | 106                 | 159  | 262  | 454  |
|                            | (6)                | (10) | (24) | (30) | (7)                 | (14) | (29) | (44) |
| Model                      | 112                | 135  | 191  | 343  | 136                 | 185  | 261  | 404  |

Default probabilities are cumulative default probabilities over 1920–2011 from Moody's investors service (2012), and credit spreads are from FISD and TRACE transaction data over 1994–2010. We report the time-series mean, with the standard deviation (reported underneath) being calculated using Newey-West procedure with 15 lags.

Table 3

Bid-ask spreads and bond-CDS spreads across credit ratings

A. Bid-ask spreads (bps)

|       | State G  |            |      | State B  |            |      |
|-------|----------|------------|------|----------|------------|------|
|       | Superior | Investment | Junk | Superior | Investment | Junk |
| Data  | 40       | 50         | 70   | 77       | 125        | 218  |
| Model | 39       | 44         | 61   | 111      | 128        | 186  |

B. Bond-CDS spreads (bps)

|       | State G |    |     |    | State B |     |     |     |
|-------|---------|----|-----|----|---------|-----|-----|-----|
|       | Aaa/Aa  | A  | Baa | Ba | Aaa/Aa  | A   | Baa | Ba  |
| Data  | 23      | 37 | 58  | 68 | 72      | 104 | 162 | 191 |
| Model | 48      | 53 | 61  | 61 | 69      | 79  | 92  | 107 |

In panel A, the normal time bid-ask spreads in the data are taken from [Edwards, Harris, and Piwowar \(2007\)](#) for median-sized trades. The numbers in recession are normal time numbers multiplied by the empirical ratio of bid-ask spread implied by Roll's measure of illiquidity (following [Bao, Pan, and Wang \(2011\)](#)) in recession time to normal time. The model-implied bid-ask spread are computed for a bond with time-to-maturity of 8 years, which is the mean time-to-maturity of frequently traded bonds (where we can compute a [Roll \(1984\)](#) measure) in the TRACE sample. The bond-CDS spreads in panel B are for 10-year bonds. When computing the model-implied bond-CDS spreads, we restrict the leverage distribution for each rating category to those firms with CDS contracts outstanding, as in the data counterpart.

## 5.4 Table 3, Figure 2, and Figure 3: liquidity moments and leverage fit

Read:

- Table 3 shows that the same model used to fit default rates and credit spreads also does a decent job on bid-ask spreads and bond-CDS spreads. This is important because these moments were not traditionally the target of structural credit models.
- Figure 2 shows the leverage distributions by rating and aggregate state. The distributions are wide, and the spreads are convex in leverage. This is why the paper averages the model over the entire empirical leverage distribution rather than evaluating it only at mean leverage.
- Figure 3 compares the model and data relation between leverage and credit spreads. The model matches this relation fairly well overall but tends to underpredict spreads for very low-leverage firms, which is consistent with the broader literature's difficulty generating high spreads for low-leverage investment-grade firms.

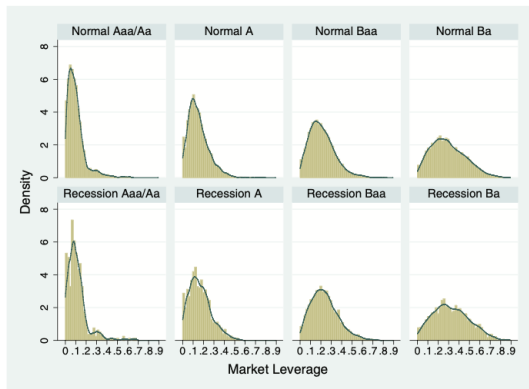


Figure 2

Empirical distribution of market leverage for Compustat firms by aggregate state and rating classes. We compute market leverage for each firm-quarter observation in Compustat from 1994 to 2012, excluding financials, utilities, and firms with zero leverage. State B is classified as quarters for which at least two months are in NBER recessions; the remaining quarters are state G. We exclude the financial crisis quarters 2008Q4 and 2009Q1.

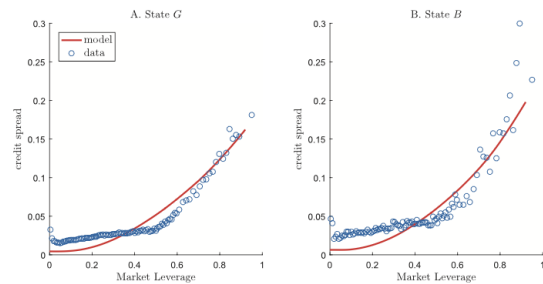


Figure 3

Credit spreads versus market leverage

This figure compares the relation between market leverage and credit spread in the model (solid line) with the data (circles).

## 5.5 Table 4: comparative statics

Read:

- The “ $h_s^c$ ” version with constant holding costs shows that price-dependent holding costs matter: without them, the cross-sectional variation in liquidity and spreads is too flat.
- Higher liquidity-shock intensities ( $\xi = 1.4, 2$ ) raise total credit spreads and bond–CDS spreads, but can lower bid-ask spreads. This is a good reminder that different liquidity measures capture different economic objects.
- Higher meeting intensities ( $\lambda = 75, 30$ ) make the market more liquid and reduce both credit spreads and bond–CDS spreads.
- Shorter average debt maturity ( $m = 1/3$ , i.e., average maturity falls from 5 to 3 years) raises rollover risk and therefore increases default probabilities and spreads. This directly validates the paper’s emphasis on rollover as the default channel through which market liquidity matters.

**Table 4**  
Comparative statics and comparison to alternative models for 10-year bonds  
A. Credit spreads (bps)

|                    | State <i>G</i> |     |     |     | State <i>B</i> |     |     |     |
|--------------------|----------------|-----|-----|-----|----------------|-----|-----|-----|
|                    | Aaa/Aa         | A   | Baa | Ba  | Aaa/Aa         | A   | Baa | Ba  |
| Benchmark          | 86             | 122 | 182 | 301 | 136            | 185 | 261 | 404 |
| $h_{C2}$           | 95             | 126 | 176 | 278 | 146            | 185 | 245 | 359 |
| $\xi = 1.4, 2$     | 106            | 137 | 192 | 313 | 165            | 209 | 282 | 432 |
| $\lambda = 75, 30$ | 65             | 87  | 130 | 232 | 108            | 139 | 193 | 313 |
| $m = 1/3$          | 96             | 141 | 210 | 332 | 148            | 209 | 296 | 445 |

B. Bid-Ask Spreads (bps)

|                    | State <i>G</i> |            |      | State <i>B</i> |            |      |
|--------------------|----------------|------------|------|----------------|------------|------|
|                    | Superior       | Investment | Junk | Superior       | Investment | Junk |
| Benchmark          | 39             | 44         | 61   | 111            | 128        | 186  |
| $h_{C2}$           | 45             | 47         | 49   | 123            | 128        | 138  |
| $\xi = 1.4, 2$     | 33             | 38         | 52   | 84             | 97         | 142  |
| $\lambda = 75, 30$ | 26             | 28         | 37   | 78             | 87         | 123  |
| $m = 1/3$          | 39             | 48         | 69   | 112            | 136        | 203  |

C. Bond-CDS Spreads (bps)

|                    | State <i>G</i> |    |     |     | State <i>B</i> |     |     |     |
|--------------------|----------------|----|-----|-----|----------------|-----|-----|-----|
|                    | Aaa/Aa         | A  | Baa | Ba  | Aaa/Aa         | A   | Baa | Ba  |
| Benchmark          | 48             | 53 | 61  | 61  | 69             | 79  | 92  | 107 |
| $h_{C2}$           | 60             | 61 | 60  | 46  | 84             | 85  | 86  | 79  |
| $\xi = 1.4, 2$     | 73             | 81 | 96  | 111 | 104            | 118 | 138 | 171 |
| $\lambda = 75, 30$ | 35             | 38 | 44  | 61  | 50             | 56  | 65  | 78  |
| $m = 1/3$          | 49             | 54 | 59  | 61  | 71             | 82  | 94  | 102 |

The “benchmark” case is our benchmark calibration. The “ $h_{C2}$ ” case is when holding costs depend on aggregate state only (we calibrate  $h_{C2} = 1.38$  and  $h_{C2} = 2.32$  to match investment-grade bid-ask spreads). The “ $\xi = 1.4, 2$ ” case is when we double the liquidity shock intensities in both states. The “ $\lambda = 75, 30$ ” case is when we increase the meeting intensity in both states from (50, 20) to (75, 30). The “ $m = 1/3$ ” case is when we lower the firm’s average debt maturity from 5 ( $m = 0.2$ ) to 3 years.

**Table 5**  
Structural decomposition for 10-year bonds across ratings

| Rating | State    | Credit spread | Structural decomposition |           |          |           |
|--------|----------|---------------|--------------------------|-----------|----------|-----------|
|        |          |               | Pure Def                 | Liq → Def | Pure Liq | Def → Liq |
| Aaa/Aa | <i>G</i> | 71            | 20                       | 2         | 45       | 5         |
|        | (%)      |               | (27)                     | (3)       | (63)     | (7)       |
|        | <i>B</i> | 96            | 22                       | 2         | 63       | 9         |
| A      | (%)      |               | (23)                     | (2)       | (66)     | (9)       |
|        | <i>G</i> | 107           | 46                       | 4         | 45       | 12        |
|        | (%)      |               | (43)                     | (4)       | (42)     | (11)      |
| Baa    | <i>B</i> | 145           | 55                       | 4         | 63       | 22        |
|        | (%)      |               | (38)                     | (3)       | (44)     | (15)      |
| Ba     | <i>G</i> | 167           | 93                       | 7         | 45       | 22        |
|        | (%)      |               | (56)                     | (4)       | (27)     | (13)      |
|        | <i>B</i> | 221           | 109                      | 9         | 63       | 41        |
| B      | (%)      |               | (49)                     | (4)       | (29)     | (18)      |
|        | <i>G</i> | 286           | 192                      | 13        | 45       | 38        |
|        | (%)      |               | (67)                     | (4)       | (16)     | (13)      |
| Ba     | <i>B</i> | 364           | 215                      | 16        | 63       | 70        |
|        | (%)      |               | (59)                     | (5)       | (17)     | (19)      |

We perform the structural liquidity–default decomposition for a 10-year bond following Section 3.1.1 given rating and aggregate state, and then aggregate over the empirical leverage distribution in Compustat. The reported credit spreads are relative to the risk-free rate, that is, net of the liquidity premium of Treasuries.

## 5.6 Table 5 and Figures 4–5: structural default–liquidity decomposition

Read:

- Table 5 is the core quantitative result. For Aaa/Aa bonds in state *G*, the total spread of 71 bps decomposes into 20 bps pure default, 2 bps liquidity-driven default, 45 bps pure liquidity, and 5 bps default-driven liquidity.
- For Ba bonds in state *B*, the total spread of 364 bps decomposes into 215 bps pure default, 16 bps liquidity-driven default, 63 bps pure liquidity, and 70 bps default-driven liquidity.
- The interaction terms are economically meaningful. Cross-sectionally, the two interaction terms together are around **10%–11%** of safe-bond spreads and **17%–24%** of junk-bond spreads depending on the macro state.
- Figure 4 visualizes the same decomposition by rating and state, making the pattern clear: safer bonds are dominated by liquidity, while riskier bonds are dominated by default, but the feedback terms are sizable throughout.
- Figure 5 turns to the time series. For lower-rated bonds, especially Ba, the default-driven liquidity component becomes large during recessions and can be as important as pure default in

explaining cyclical spread movements.

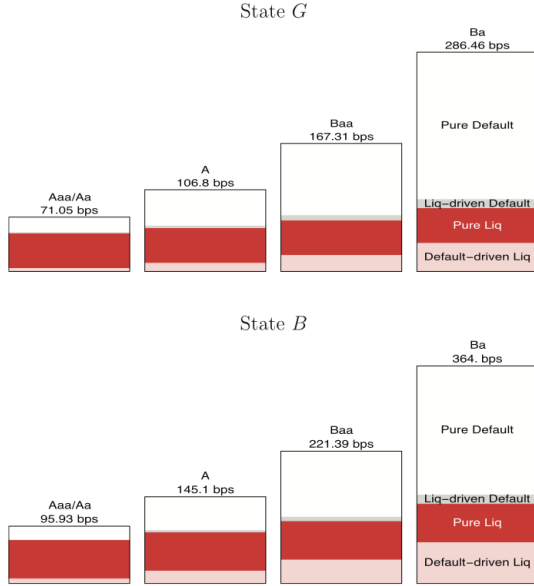


Figure 4  
Graphical illustrations of structural liquidity-default decomposition for 10-year bonds across ratings  
For numbers and explanations, see Table 5

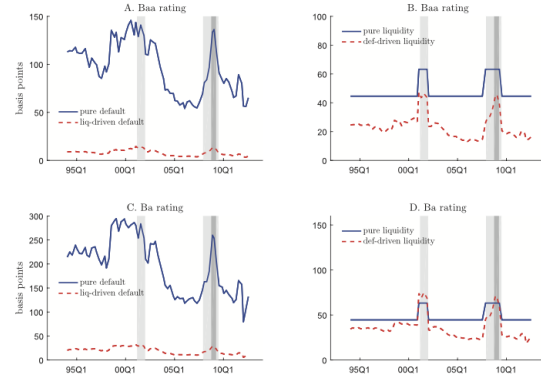
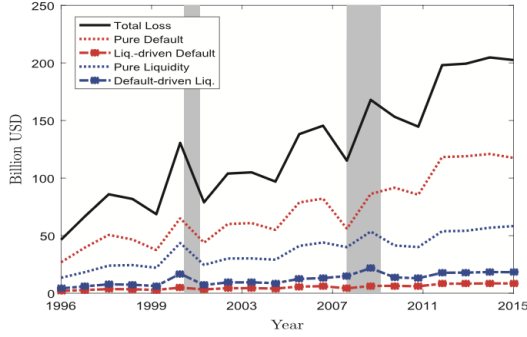


Figure 5  
Time-series structural decomposition of credit spreads for Baa- and Ba-rated firms  
For each firm-quarter observation, we locate the corresponding cashflow level  $y$  that delivers the observed market leverage in Compustat (excluding financial and utility firms) and perform the structural liquidity-default decomposition for a 10-year bond following the procedure discussed in Section 3.1. For a given credit rating (Baa or Ba), we average across firms to obtain each component for each quarter from 1994 to 2012. Recessions are highlighted in gray. For completeness, we also calculate the model-implied decomposition results for the crisis period from 2008Q4 to 2009Q1 in dark gray (which is excluded from the rest of this paper).

## 5.7 Figure 6 and Table 6: aggregate losses and the liquidity-policy counterfactual

Read:

- Figure 6 translates spread decompositions into dollars using annual issuance. Total losses for new issuance from 1996 to 2015 amount to about **\$2.9 trillion**, or **14%** of total issuance.
- Of this, pure default accounts for about **\$1.65 trillion** (56.8%), liquidity-driven default for about **\$0.12 trillion** (4.1%), pure liquidity for about **\$0.86 trillion** (29.4%), and default-driven liquidity for about **\$0.28 trillion** (9.6%).
- Table 6 studies a recession liquidity intervention. For Ba-rated bonds, spreads fall from **364** to **262** bps in state *B*, and from **286** to **234** bps even in state *G* because better recession liquidity changes forward-looking default incentives.
- The decomposition of the policy effect is especially important. For Ba bonds in state *B*, only **42%** of the spread reduction comes from the direct pure-liquidity channel, while **9%** comes from liquidity-driven default and **49%** from default-driven liquidity.
- The dollar value of the same policy is large: it raises aggregate corporate bond market value by roughly **\$256 billion**. Ignoring default-liquidity interactions would understate this effect, giving only about **\$173 billion**.



**Figure 6**  
Annual total loss in value for new issuances in the corporate bond market and its decomposition  
This figure plots our estimates of the total loss in value for new issuances in the corporate bond market due to default risk and liquidity frictions. At bond level, the loss in value is measured as the gap between the price of a defaultable bond and the price of the same bond if it is default-free and not subject to liquidity frictions.

**Table 6**  
Effect of liquidity provision policy on 10-year bonds across ratings

| Rating | State | Credit spread |           | Contribution of each component |             |             |
|--------|-------|---------------|-----------|--------------------------------|-------------|-------------|
|        |       | w/o policy    | w/ policy | pure LIQ (%)                   | LIQ→DEF (%) | DEF→LIQ (%) |
| Aaa/Aa | G     | 71.1          | 41.9      | 83                             | 5           | 12          |
|        | B     | 96.0          | 44.4      | 83                             | 3           | 14          |
| A      | G     | 107           | 71.9      | 69                             | 8           | 23          |
|        | B     | 145           | 82.1      | 68                             | 5           | 27          |
| Baa    | G     | 167           | 125       | 57                             | 10          | 33          |
|        | B     | 221           | 143       | 55                             | 7           | 38          |
| Ba     | G     | 286           | 234       | 46                             | 12          | 42          |
|        | B     | 364           | 262       | 42                             | 9           | 49          |

We consider a policy experiment that improves the liquidity environment ( $\chi$  and  $\lambda$ ) in the  $B$  state to be as good as  $G$  state (i.e.,  $\chi_B = 0.06$  and  $\lambda_B = 50$ ). We fix the distribution of cash flow levels  $y$  at the values that deliver the observed market leverage distribution in Compustat (excluding financial and utility firms) for the corresponding state in our baseline calibration. We then report the average credit spreads (relative to the risk-free rate) under the policy for each state together with credit spread without policy. We perform the structural liquidity-default decomposition to examine the channels that are responsible for the reduced borrowing cost. We report the percentage contribution of each component to the credit spread change.

## 6 Short summary

- **Method.** The paper builds a structural credit model with endogenous default, debt rollover, OTC secondary-market trading, and bond-price-dependent holding costs.
- **Identification/calibration.** It matches firm default rates, credit spreads, bid-ask spreads, and bond-CDS spreads using a small set of calibrated liquidity parameters and the empirical leverage distribution of firms.
- **Main result.** Liquidity and default are not additively separable. Their interaction explains a meaningful share of both the level and cyclical movement of credit spreads.
- **Policy implication.** Liquidity provision can lower spreads not only directly, but also indirectly by reducing default risk and by improving liquidity further as bonds become safer.

## 7 Conclusion

### 7.1 Core conclusion (what the paper establishes)

- The paper shows that a structural credit model with rollover risk and endogenous bond-market liquidity can jointly explain default-related and liquidity-related bond-pricing moments.
- It also shows that the interaction between default and liquidity is quantitatively important, both in the cross section and over the business cycle.
- Finally, it demonstrates that ignoring this interaction understates the financing costs of illiquidity and the potential benefits of liquidity policy.

### 7.2 Economic intuition

- A bond that becomes harder to sell is less valuable to investors, especially when they face liquidity shocks.
- Lower bond prices make refinancing more expensive for firms with maturing debt.
- More expensive refinancing raises default incentives for equityholders.



- Higher default risk then feeds back into lower bond prices and worse liquidity because the bond becomes a worse store of value and collateral asset.
- This is why the paper's decomposition produces two interaction terms, not just separate default and liquidity pieces.

### **7.3 Takeaway**

This paper bridges structural credit risk and market liquidity in a quantitative framework. Its main message is that bond market liquidity is not merely a residual component of credit spreads: it can change default behavior through rollover risk, and that interaction materially affects spreads, firm financing costs, and the value of policy interventions. Any analysis of corporate bond pricing or liquidity support that ignores this feedback will miss an important part of the economics.